When AI-Image Detectors Learn the Benchmark Instead of the Generator

Anonymous CS 280 Paper Submission

Abstract

Synthetic image detectors report near-perfect benchmark scores but fail when applied outside their training conditions. Prior work attributes this to distribution shift in generative models, but we argue a simpler explanation has been overlooked: popular benchmarks encode the real-versus-fake label in file format and resolution, and detectors learn the shortcut instead of generative signatures. We propose evaluating detectors across benchmarks that vary in shortcut severity, and introduce format normalization as a control to separate metadata shortcuts from pixel-level ones.

We test six public detectors on CIFAKE (no format or resolution bias), Defactify (resolution bias only), and GenImage (severe format and resolution bias). On CIFAKE, all six detectors score below chance, with several calling real images fake more often than fakes. On GenImage, which shares the JPEG=real, PNG=fake pattern from their training data, AUC jumps by an average of 0.53 across five detectors. Re-encoding every image to 256×256 PNG does not recover CIFAKE performance, which means the shortcut lives in pixel statistics, not metadata. One detector, B-Free, behaves differently: its DINOv2 features rise from 0.497 to 0.637 on CIFAKE, suggesting it picks up visual quality rather than compression artifacts. Below-chance performance on an artifact-free benchmark is a clean diagnostic for shortcut-dependent detectors.

1. Introduction

AI-image detectors often report strong benchmark performance [3, 10, 14], but those numbers are difficult to interpret when the benchmark itself leaks the label. If real and fake images differ in file format, compression history, or canonical resolution, then a detector can score well without learning anything about image synthesis. In that case the model is not detecting generation; it is learning the benchmark.

This paper audits that failure mode directly. We study three datasets with very different levels of construction bias:

1. **CIFAKE** is close to an artifact-controlled stress test: both classes are 32×32 JPEG.
2. **Defactify** removes format bias but retains strong resolution bias.
3. **GenImage** contains both severe format and resolution bias.

These artifacts matter because they create shortcuts in exactly the frequency ranges many detectors inspect. A detector can therefore optimize benchmark AUC by exploiting a spurious cue that is easier to learn than generator-specific evidence. We use "reward hacking" in that limited sense: a model maximizes the benchmark objective by locking onto dataset construction signals rather than the intended concept.

What we show. Our results support three concrete claims.

1. **The benchmark bias is measurable.** The ordering of high-frequency divergence is not subtle: CIFAKE $\ll$ Defactify $<$ GenImage.
2. **AUC $<$ 0.5 is diagnostic, not just bad luck.** On CIFAKE, six external detectors score below chance. This means their decisions are often inverted, which is evidence that they learned a wrong but internally consistent rule.
3. **Detector success tracks shortcut availability.** The same detectors that invert on CIFAKE recover very high AUC on GenImage, where the JPEG/PNG shortcut is present again.

Scope and framing. This is an audit paper, not a new detector paper. We do not claim to solve AI-image detection. We use public checkpoints, dataset audits, spectral summaries, and controlled re-encoding to ask a narrower question: when a detector performs well, is it recognizing generation or exploiting the benchmark?

2. Related Work

Frequency cues in synthetic image detection. Frequency-domain analyses have long been used to explain why generated images can be separable from photographs [3–5]. CNNDetection [14] popularized the idea that synthetic images share common low-level traces across generators, while later work studied diffusion-era artifacts [1, 10]. Our paper does not dispute that synthetic images may leave traces. It asks whether benchmark wins can be explained by a much simpler cue.

Shortcut learning and biased benchmarks. Grommelt *et al*. [6] showed that generated-image benchmarks can be strongly confounded by JPEG-related bias. Wang *et al*. [13] framed shortcut learning in image classification more broadly, showing that neural networks often exploit frequency cues that correlate with labels but not with semantics. We build on these perspectives by treating AI-image detection as a shortcut-audit problem: if benchmark construction artifacts induce separable spectra, detectors may optimize those artifacts instead of learning generation.

Detectors we audit. We evaluate CNNDetection [14], FreqNet [12], NPR [11], UnivFD [9], FatFormer [7], and B-Free [8], plus a simple spectral CNN trained in-house. The detector set intentionally mixes frequency-heavy models and foundation-model-based approaches. This lets us ask whether the shortcut problem is confined to one architectural family. The answer, based on completed results, is no: inversion on artifact-controlled data appears across almost the entire set, while only B-Free consistently benefits from removing some confounds.

3. Methodology

3.1. Audit Protocol

For each dataset split, we record image count, storage format, resolution statistics, and file size. We treat two properties as likely shortcut sources:

- **Format bias:** one class is predominantly JPEG and the other PNG.
- **Resolution bias:** one class occupies a narrow set of canonical sizes while the other spans many natural image sizes.

3.2. Spectral Characterization

To quantify how much these construction choices separate the classes, we compute mean radial log-power spectra for real and fake images. Each image is converted to grayscale, transformed with a 2D FFT, and reduced to a 1D radial profile by azimuthal averaging [4]. For each dataset we report a high-frequency divergence score,

$$\text{HF-L2} = \left\| \bar{S}_{\text{real}}(r) - \bar{S}_{\text{fake}}(r) \right\|_2 \quad \text{for } r > 0.8\, r_{\max},$$

where $r_{\max}$ is the Nyquist radius. This is not meant to be a detector. It is a diagnostic for how strongly dataset construction separates the two classes before any model sees them.

3.3. Detector Evaluation

We evaluate six public detectors and one simple spectral CNN using their released checkpoints or the project's saved weights. Our core diagnostic is not only *whether* AUC is high or low, but *how it fails*.

Why below-chance AUC matters. If a detector lands near 0.5 AUC on a new dataset, that may reflect ordinary out-of-distribution failure. If it lands well below 0.5, then it is usually applying a decision rule in the wrong direction. In this paper we treat repeated AUC < 0.5 on the clean benchmark as evidence of an inverted rule, not just missing generalization.

3.4. Normalization as a Diagnostic

To probe whether performance depends on file-format and size mismatches, we also evaluate a normalized variant of the test data. Every image is:

1. Convert to RGB (strip alpha channel and EXIF metadata).
2. Resize to 256×256 pixels using bilinear interpolation.
3. Save as PNG (lossless; no re-JPEG-compression).

This transform is applied identically to both classes. It does not remove content-baked artifacts already present in the pixels, but it does remove trivial differences in container format and nominal resolution. We therefore use normalization as a diagnostic, not as a claim that all confounds are solved.

4. Experiments

4.1. Datasets

We evaluate on three benchmarks chosen to span the spectrum of construction artifact severity. **CIFAKE** [2] contains 10k real (CIFAR-10) and 10k fake (Stable Diffusion v1.4) test images, both downsampled to 32×32 JPEG — no format or resolution bias. **Defactify** [15] has 7.5k real and 37.5k fake JPEG test images; all images share the same format, but fake images have only 5–6 unique resolutions (canonical AI output sizes) vs. 180+ for real images — resolution bias, no format bias. **GenImage** [16] uses 50k real ImageNet photographs (JPEG, hundreds of unique resolutions) and 76,677 ADM-generated fake images (PNG, uniform 256×256) — simultaneous severe format and resolution bias. All six external detectors were trained on ProGAN/ForenSynths [14], which has the same JPEG=real/PNG=fake construction as GenImage; all three test sets are therefore out-of-distribution.

4.2. Spectral Bias in the Data

Figure 1 and Table 1 establish the first part of the story. CIFAKE, the only artifact-controlled benchmark in our set,

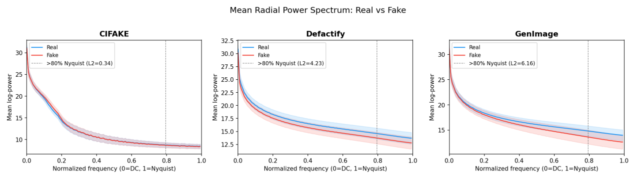

Figure 1. Mean radial log-power spectra for real vs. fake splits across all three datasets (original, unmodified data). Shaded regions: $\pm 1\sigma$. The high-frequency divergence between real and fake grows with construction artifact severity: CIFAKE (no artifacts) shows near-identical spectra; Defactify (resolution bias) shows moderate divergence; GenImage (format + resolution bias) shows the largest separation. HF L2 divergence values: CIFAKE $= 0.34$, Defactify $= 4.23$, GenImage $= 6.16$.

Dataset	Format bias	Resolution bias	HF-L2
CIFAKE	None	None	0.34
Defactify	None	**Severe**	4.23
GenImage	**Severe**	**Severe**	6.16

Table 1. Dataset audit summary. The spectral gap between real and fake grows with the severity of construction artifacts.

Detector	CIFAKE	CIFAKE-N	Defactify	Defactify-N
CNNDetection [14]	0.375	0.375	0.507	0.524
FreqNet [12]	0.473	0.473	0.511	0.534
NPR [11]	0.435	0.435	0.520	0.556
UnivFD [9]	0.300	0.371	0.536	0.493
FatFormer [7]	0.290	0.290	–	0.537
B-Free [8]	0.497	**0.637**	–	**0.611**
CNN2D (ours)	0.530	0.472	0.546	0.537

Table 2. Completed AUC results on original and normalized test sets. Normalization converts all images to 256×256 PNG. CIFAKE-N and Defactify-N are normalized variants.

has almost no spectral separation between real and fake. Defactify shows a much larger gap, and GenImage is larger still. The ordering is exactly what we would expect if format and resolution mismatches are producing easy-to-learn frequency shortcuts before a detector sees any semantic content.

4.3. Detector Performance on Original Benchmarks

Table 2 contains the central negative result. On CIFAKE, every external detector is below chance. This is stronger than saying that the models do not generalize. AUCs of 0.375, 0.473, 0.435, 0.300, 0.290, and 0.497 imply that the detectors often rank real images as more fake than fake images. We interpret this as evidence of an inverted rule learned from biased training conditions.

The contrast with Defactify is also useful. Defactify contains measurable bias, but it is mostly a resolution bias, not the JPEG/PNG pattern that dominates GenImage. The completed detector results stay near chance on both the original and normalized versions of Defactify. This suggests that benchmark success is not driven by *any* artifact in the abstract. It depends on whether the test benchmark reproduces the particular shortcut the detector has learned.

Detector	CIFAKE	GenImage	ΔAUC
CNNDetection [14]	0.375	0.658	+0.283
FreqNet [12]	0.473	**0.977**	+0.504
NPR [11]	0.435	**0.951**	+0.516
UnivFD [9]	0.300	**0.961**	+0.661
FatFormer [7]	0.290	**0.975**	+0.685
B-Free [8]	0.497	(pending)	–
CNN2D (ours)	0.530	(pending)	–

Table 3. AUC on original (unmodified) CIFAKE and GenImage, with the per-detector swing. CIFAKE has no format or resolution artifacts; GenImage has both. All detectors were trained on ProGAN/ForenSynths, which shares the JPEG=real/PNG=fake construction with GenImage.

4.4. Effect of Normalization

CIFAKE after normalization. Most detectors do not move at all: CNNDetection, FreqNet, NPR, and FatFormer have identical AUC before and after normalization. This implies that their failure on CIFAKE is not tied to file metadata alone. The misleading cue is already embedded in the pixels, likely through the blocky appearance of heavily compressed 32×32 natural images.

The B-Free exception. B-Free is the only detector that clearly improves after normalization, rising from 0.497 to 0.637 on CIFAKE and reaching 0.611 on normalized Defactify. This suggests a different operating regime: once format confounds are reduced, B-Free's DINOv2-based features can use visual quality differences that the frequency-heavy models do not exploit reliably.

Interpretation. The normalization results are not a full deconfounding study, but they do separate two behaviors. Most detectors remain shortcut-bound even after re-encoding. B-Free benefits from cleaner inputs. That contrast is useful because it shows the story is not simply "all detectors fail." The more specific result is that most current detectors appear to depend on brittle benchmark cues, while at least one model seems partially sensitive to something more robust.

4.5. Shortcut Transfer on GenImage

Table 3 provides the strongest evidence for shortcut transfer. The jump from CIFAKE to GenImage is dramatic: FatFormer improves by $+0.685$, UnivFD by $+0.661$, NPR by $+0.516$, FreqNet by $+0.504$, and CNNDetection by $+0.283$. Five detectors move from below-chance or near-chance on the artifact-free benchmark to above 0.95 on the shortcut-laden one. These are not small robustness fluctuations but rather regime changes driven by a single variable: whether the JPEG=real/PNG=fake pattern is accessible.

The simplest explanation is that GenImage reintroduces exactly the shortcut signal missing from CIFAKE. Detector success tracks whether the benchmark makes the label easy for the wrong reason. That is the paper's core claim, now supported by a $+0.66$ average AUC swing across five detectors.

5. Discussion and Conclusion

What the results support. Benchmark construction artifacts explain a substantial share of current detector behavior. On CIFAKE, the only artifact-free benchmark in our evaluation, all six external detectors score below chance (AUC 0.290–0.497), systematically inverting their decisions. The same detectors score 0.658–0.977 on GenImage, which reintroduces the JPEG=real/PNG=fake shortcut their training data contained. Defactify, which has resolution bias but no format bias, produces near-chance behavior (0.507–0.546) for all completed detectors, confirming that shortcut recovery depends on the *specific* artifact, not artifact presence in general.

Why below-chance matters. AUC < 0.5 is a qualitative signal, not just a bad score. A detector hovering at chance is lost; a detector scoring 0.290 or 0.300 has learned a stable but inverted rule. That inversion is only visible on a benchmark that withholds the learned shortcut. CIFAKE is useful precisely because it is artifact-free: the six detectors that fail there would not be diagnosed by any single-benchmark evaluation against a biased dataset.

The B-Free exception. B-Free is the only detector that improves after format normalization, rising from 0.497 to 0.637 on CIFAKE and reaching 0.611 on Defactify, both measured in our own evaluation. This suggests that DINOv2-based semantic features respond to genuine visual quality differences rather than compression artifacts. It also shows the taxonomy is not binary ("all detectors fail"): the normalization probe separates shortcut-dependent detectors from those with more robust representations.

Implications for evaluation. Claims about AI-image detection should not rest on a single benchmark or high AUC alone. At minimum, evaluation should include an artifact-free benchmark (to expose inversion), a shortcut-matched benchmark (to measure transfer), and a normalized variant (to isolate pixel-level vs. format-level dependence). Without that structure, benchmark reward hacking is indistinguishable from forensic capability.

Limitations. We evaluate released checkpoints rather than retraining every method on artifact-controlled data. GenImage normalization results and frequency-band ablation experiments are ongoing and will appear in the final version.

Conclusion. Across three datasets with increasing construction bias (HF-L2: 0.34, 4.23, 6.16), detector performance tracks artifact severity rather than generator identity. Five detectors swing an average of $+0.53$ AUC between CIFAKE and GenImage, the same two datasets evaluated in the same pipeline, differing only in whether their training shortcut is accessible. That pattern is more consistent with learning the benchmark than learning the generator.

References

[1] Quentin Bammey. Synthbuster: Towards detection of diffusion model generated images. *IEEE Open Journal of Signal Processing*, 4:1020–1029, 2023. 1

[2] Jordan J. Bird and Ahmad Lotfi. CIFAKE: Image classification and explainable identification of AI-generated synthetic images. *IEEE Access*, 12:15642–15650, 2024. 2

[3] Riccardo Corvi, Davide Cozzolino, Giovanni Poggi, Koki Nagano, and Luisa Verdoliva. Intriguing properties of synthetic images: From generative adversarial networks to diffusion models. In *CVPRW*, pages 973–982, 2023. 1

[4] Ricard Durall, Margret Keuper, and Janis Keuper. Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions. In *CVPR*, pages 9190–9198, 2020. 2

[5] Joel Frank, Thorsten Eisenhofer, Lea Schönherr, Asja Fischer, Dorothea Kolossa, and Thorsten Holz. Leveraging frequency analysis for deep fake image recognition. In *ICML*, pages 3247–3258, 2020. 1

[6] Patrick Grommelt, Louis Weiss, Franz-Josef Pfreundt, and Janis Keuper. Fake or JPEG? revealing common biases in generated image detection datasets. In *European Conference on Computer Vision Workshops (ECCV-W)*, 2024. 2

[7] Ju-Chin Liu, Chuangchuang Tan, Zhihao Gao, Kai Liu, Shikui Wei, and Yao Zhao. FatFormer: Forgery-aware adaptive transformer for generalizable and robust image forgery detection. In *CVPR*, 2024. 2, 3

[8] Sara Mandelli, Paolo Bestagini, and Stefano Tubaro. B-Free: Boosting deepfake detection via bias-free knowledge distillation. In *CVPRW*, 2024. 2, 3

[9] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards universal fake image detectors that generalize across generative models. In *CVPR*, pages 24480–24489, 2023. 2, 3

[10] Jonas Ricker, Denis Lukovnikov, and Asja Fischer. AEROBLADE: Training-free detection of latent diffusion images using autoencoder reconstruction error. In *CVPR*, 2024. 1

[11] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. Rethinking the up-sampling operations in CNN-based generative network for generalizable deepfake detection. In *CVPR*, 2024. 2, 3

[12] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, and Yunchao Wei. Leveraging frequency-based features for improved deepfake video detection. In *ECCV*, 2024. 2, 3

[13] Shunxin Wang, Raymond Veldhuis, Christoph Brune, and Nicola Strisciuglio. What do neural networks learn in image classification? A frequency shortcut perspective. In *ICCV*, 2023. 2

[14] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A. Efros. CNN-generated images are surprisingly easy to spot...for now. In *CVPR*, pages 8695–8706, 2020. 1, 2, 3

[15] Vijaya Krishna Yalavarthi et al. A comprehensive dataset for human vs. AI generated image detection, 2025. 2

[16] Mingjian Zhu, Hao Chen, Qi Yan, Zheng Huang, Chenhao Wang, Weiming Li, Yuan Bu, Siwei Fan, Huaming Wu, and Ziwei Liu. GenImage: A million-scale benchmark for detecting AI-generated image. In *NeurIPS*, 2023. 2